

Perceived isolation, social disconnectedness, and the mental health of young adults in South Korea

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Abstract

Background

While perceived isolation and social disconnectedness in older adults and adolescents are considered to be risk factors for mental health, limited research on the association between these variables in young adults. This study aims to investigate the prevalence of perceived isolation and social disconnectedness among young adults in Korea and the impact of these factors on mental health.

Methods

We utilized data from the 2021 Young Adult Socio-economic Survey, which approved by Statistics Korea. A sample of 2,041 participants was selected. Logistic regression analysis was conducted to examine the association between perceived isolation and social disconnectedness and mental health.

Results

Perceived isolation and social disconnectedness were associated with mental health. Perceived isolation (OR 1.86, 95% CI = 1.61–2.15) and going out (OR 2.59, 95% CI = 1.47–4.58) significantly influenced depression. Additionally, subjective well-being was significantly influenced by perceived isolation (OR 1.17, 95% CI = 1.03–1.33), going out (OR 1.96 95% CI = 1.21–3.16), and number of friends (OR 0.79, 95% CI = 0.69–0.90).

Conclusions

Overall, the findings suggest that social connection is crucial to young adults' mental health.

Introduction

Social isolation refers to an individual's lack of social contact and satisfaction with their relationships with other individuals [1]. Within the construct of social isolation, there are two related but distinct concepts: perceived isolation and social disconnectedness [2]. Perceived isolation and social disconnectedness are significant public health concerns that could have adverse effects on an individual's physical health, mental health, and quality of life throughout their lifespan [3]; various studies have demonstrated that social isolation is a risk factor for mortality, chronic disease, and subjective health, including perceived isolation and social disconnection [4–6]. Studies on the link between social isolation and healthcare spending have shown that Medicare spends \$1643 more annual per capita on socially isolated people [7]. Prior research has predominantly focused on examining the association between individual aspects of social isolation and mental health, such as perceived isolation and mental health [8–10] or social disconnection and mental health [11, 12]. Therefore, there is a dearth of research

that comprehensively investigates the joint influence of both social isolation and social disconnection on mental health outcomes. This research aims to investigate the relationship between perceived isolation, social disconnection which are two aspects of social isolation and mental health.

Perceived isolation is a subjective experience of loneliness, a lack of social connection and support, and a perceived absence of intimacy in relationships with others [13]. It is measured by assessing the emotional and instrumental support an individual receives from their family members, friends, and spouse or partner, as well as indicators such as companionship, loneliness, and feelings of being isolated [14]. By contrast, social disconnectedness primarily refers to a quantitative lack of social contact with others [15], and is measured by various indicators, such as social network size, social network coverage, frequency of interaction with network members, percentage of network members residing in the same household, number of friends, attendance at group meetings, frequency of socializing with friends and family, and engagement in volunteer activity [16]. It is worth noting that an individual can experience feelings of isolation even with a high level of social contact, while someone with minimal social contact may not feel isolated [2, 14]. In other words, although perceived isolation and social disconnectedness have overlapping aspects, they also exhibit conceptual differences [2, 17]. Therefore, understanding the complex relationship between perceived isolation and social disconnectedness necessitates considering both factors.

Perceived isolation and social disconnectedness are recognized as risk factors for poor health outcomes. Health is influenced by interactions with various social environments, and social connection has been identified as a determinant of an individual's overall health [18, 19]. For instance, individuals reporting higher levels of perceived isolation have been found to have a 1.19 times greater likelihood of experiencing depression and a 1.12 times greater likelihood of experiencing anxiety, independent of age, gender, partnership status, and socioeconomic factors [20]. Moreover, individuals perceiving themselves as lonely are 41% more likely to have chronic diseases, 40% more likely to have diabetes, and 94% more likely to report worsening subjective health [4]. Similarly, social disconnectedness, characterized by a lack of social contact with friends and relatives, has been linked to reduced health-related quality of life [5]. Both perceived social isolation and social disconnectedness have been associated with increased incidence of psychological, cognitive, and physical illnesses [2, 21]. Notably, the two dimensions of social isolation have been shown to increase the risk of coronary heart disease by approximately 29%, stroke by 32%, and dementia by 50%, as well as accelerate cognitive decline [22–24]. Additionally, Schwarzbach et al.'s (2014) systematic literature review of 10 years of research revealed that the less social relationships and social contacts individuals have, the more depressive symptoms they suffer. This emphasizes the importance of supporting social relationships and mental health [25].

Recent evidence suggests that young adults 18 ~ 30 years old are particularly vulnerable to perceived isolation and social disconnectedness compared to other age group [26, 27]. This population undergoes significant life transitions when entering college, getting married, and starting a new job, which can impact their social relationships [28]. Consequently, young adults may experience feelings of isolation as they navigate new social environments and take on new social roles [2]. In South Korea, young adults

face additional risks of voluntary or situational isolation due to socio-economic challenges such as high unemployment rates, student loan debts, and rising housing costs [29]. Moreover, the recent rise in single-person households has further contributed to this risk [30]. Notably, COVID-19 pandemic and the subsequent social distancing measures have exacerbated social isolation among young individuals [31]. Therefore, social isolation among young adults in South Korea has emerged as a pressing social issue. A survey conducted by the National Youth Policy Institute in 2018 revealed that 13.4% of young adults aged 19–34 in South Korea reported feelings of isolation from others, while 16.6% expressed feeling a sense of complete loneliness [32].

Young adults in Korea are facing a concerning rise in depression rates, which has resulted in a significant increase in their hospital visits for depression. According to statistics from the Health Insurance Review & Assessment Service, young adults in Korean society experience the highest levels of depression and visit the hospital for treatment of depression [33]. As of 2021, the number of patients hospitalized with depression was highest among people aged 20–39 among a total of 910,785 people; the number of patients in their 20s increased by about 8%, representing the largest increase among all age groups [33]. Furthermore, young adults accounted for approximately one-third of all diagnosed cases of depression [33]. These statistics suggest that considerable attention should be directed toward the mental health of young adults in Korea. The prevalence of suicidal thoughts closely associated with depression was also alarmingly high among young adults, with a rate of 22.9% [34]. South Korea has consistently held the highest suicide rate among the Organisation for Economic Co-operation and Development countries since 2003, with the younger population aged 10–39 being particularly affected by this trend [35].

Previous studies have demonstrated that elevated levels of perceived isolation and social disconnectedness among adolescents and older adults are significantly associated with increased depression [2, 36]. Whereas the link between perceived isolation, social disconnectedness, and mental health has been well-established for older adults [26], the exploration of this connection for young adults remains relatively limited. Consequently, further investigation into the association between social isolation and mental health outcomes in young adults is warranted [37]. In addition, since most research related to this topic has been conducted with US or European populations [20], and few studies have specifically examined this relationship among young adults in Korea, further research is needed. Moreover, previous studies often failed to consider both perceived isolation and social disconnectedness concurrently when examining social isolation. Additionally, research on social isolation has predominantly focused on older adults experiencing social disconnectedness due to aging [38–40], with relatively few studies investigating the case of young adults. Given that the increased isolation experienced by young adults and the prevalence of depression are pressing societal issues in Korea, exploring the relationship between social isolation and mental health outcomes in this population is crucial. Therefore, this study aimed to investigate the impact of perceived isolation and social disconnectedness on depression and quality of life among Korean young adults aged 19–34.

Methods

Data

This study utilizes data from the Young Adult Socio-economic Survey an annual survey conducted to gain insights into the educational, marital, reproductive, and employment status of young adults. The 2021 survey specifically focuses on individuals aged 18 to 34 and provides comprehensive information on their socio-economic status, health and well-being, and social welfare needs, in accordance with the FRAMEWORK ACT ON YOUTH in Korea. Additionally, a supplementary survey aims to determine the prevalence and characteristics of social isolation [32].

The survey areas used for the sampling design of the Young Adult Socio-Economic Survey, were based on the 2018 General Population and Housing Survey conducted by Statistics Korea. A total of 290 sampling units were selected in the first stage using a two-stage stratified sampling method. In the second stage, households and young adults residing in these survey areas were selected based on the proportion of sampling units in each stratum. The survey was conducted over a period of approximately 15 weeks, from July 19 to October 31, 2021. A total of 2,041 young adults participated in the survey, and detailed information was collected through one-on-one, face-to-face interviews and TAPI (Tablet Assisted Personal Interviews) utilizing tablet PCs [32].

Measures

Perceived isolation

Perceived isolation was assessed using a translated version of the 3-item Loneliness Scale developed from the UCLA Loneliness Scale (version 3), a widely used measure of loneliness. The 3-item Loneliness Scale was previously validated and demonstrated adequate validity [2, 41, 42, 43]. The scale comprises the following three items: "1. How often do you feel that you lack companionship?", "2. How often do you feel left out?", and "3. How often do you feel isolated from others?". Participants responded to each item using a 3-point Likert scale, ranging from 1 for "hardly ever or never" to 3 for "often." The total score, ranging from 3 to 9, was calculated by summing the responses across all items. Higher scores indicate higher levels of perceived social isolation. The internal consistency of the scale in this study was assessed using Cronbach's alpha coefficient, which yielded a value of 0.643, indicating acceptable internal consistency.

Social disconnectedness

Social disconnectedness has been assessed through various measures in prior research. In this study, we employed two indicators: frequency of going out and number of friends. Frequency of going out was assessed using a single item that inquired how often an individual typically engages in activities outside their residence. The response options included going out every day, 3–4 days a week, often, and sometimes for purposes such as work, school, leisure, or socializing. Individuals who fell into any of these categories were considered as "going out". Conversely, individuals who primarily stayed at home but occasionally ventured out for local errands, such as visiting a nearby convenience store, and those

who never left their houses, were classified as "not going out." The number of friends was categorized based on the following criteria derived from previous research: 0 for no friends, 1 for one friend, 2 for 2–3 friends, 3 for 4–9 friends, and 4 for 10 or more friends. This categorization scheme allowed for a quantification of individuals' social network size. By employing these indicators, we aimed to capture different facets of social disconnectedness among participants in our study.

Depression

Depressive symptoms were assessed using the Center for Epidemiological Studies-Depression 11 (CES-D 11) scale, which includes 11 items measuring participants' feelings over the past two weeks. The items encompass a range of experiences, such as loss of appetite ("I don't want to eat and have no appetite"), overall well-being ("I've been doing relatively well"), and depressed mood ("I've been quite depressed"), among others. Each item is scored on a scale from 0 to 3, indicating the extent to which participants experienced each symptom. To calculate the total score, the individual item scores were summed. The sum was then multiplied by 20/11, as per the scoring guidelines of the CES-D 11, to yield the total depression score. A score of 16 or higher on the scale was considered indicative of depressive symptoms [44, 45]. The internal consistency of the CES-D 11 scale in this study was evaluated using Cronbach's alpha coefficient, which yielded a value of 0.855. This indicates high internal consistency, suggesting that the scale items measure the same underlying construct of depression in a reliable manner.

Subjective well-being

Subjective well-being was assessed using the 5-item World Health Organization Well-Being Index (WHO-5) [46, 47], a widely recognized measure employed by the World Health Organization to gauge individuals' subjective well-being. The WHO-5 comprises five items that capture participants' self-reported well-being over the past two weeks. The five items of the WHO-5 are as follows: "I have felt cheerful and in good spirits," "I have felt calm and relaxed," "I have felt active and vigorous," "I woke up feeling fresh and rested," and "My daily life has been filled with things that interest me." Each item is rated on a scale ranging from 0 to 5 points, allowing participants to indicate the degree to which they have experienced each aspect of well-being. To calculate the total well-being score, the individual item scores were summed and then multiplied by 4, as recommended in the scoring guidelines for the WHO-5. This transformation yields a total score ranging from 0 to 100, with higher scores indicating greater subjective well-being. A score below 50 was considered indicative of poor well-being [46]. In this study, the internal consistency of the WHO-5 scale was assessed using Cronbach's alpha coefficient, yielding a value of 0.897. This high value indicates excellent internal consistency, suggesting that the scale items measure the same underlying construct of subjective well-being in a reliable manner.

Demographic information

Demographic information was collected, including gender, age, education, income, marital status, living arrangements, economic activity, and self-rated health. Gender was included as a categorical variable, with the reference category being male. Age was measured in years. Education was assessed using three categories: high school graduate or less, community college graduate, and four-year college graduate,

with the reference category being high school graduate or less. Income was measured as a continuous variable. Marital status was categorized as either having no spouse (reference category) or being married or in a committed relationship. Living arrangements were categorized as living alone (reference category), living with parents, or living with a partner/spouse. Economic activity was included as a categorical variable, with the reference category being engaged in economic activity. Self-rated health was assessed using a 5-point scale ranging from 1 (very poor) to 5 (very good). Poor health was defined as a score of less than 3 on the self-rated health scale. These demographic variables were collected to provide a comprehensive understanding of the study participants' characteristics and enable the examination of potential associations and enable the examination of potential associations with perceived isolation, social disconnectedness, and mental health outcomes.

Data Analysis

Descriptive statistics were conducted to provide an overview of the participants' general characteristics. Multiple logistic regression analyses were employed to examine the association between perceived isolation, social disconnectedness, and mental health outcomes. This study did not require institutional review board approval because the data from Young Adult Socio-economic Survey are publicly available. All statistical analyses were performed using STATA version 16.1.

Results

Table 1 presents the descriptive statistics of participants' perceived isolation, frequency of going out, number of friends, depression, subjective well-being, and demographic characteristics. A total of 2,041 participants, consisting of 1,074 men and 967 women, with a mean age of 26.2 years, were included in the study. Using The UCLA 3-item Loneliness Scale, we found that on average, participants reported a perceived social isolation score of 3.4. This indicates that the majority of young adults in the sample reported relatively low levels of perceived social isolation, while a small percentage of young adults did indicate experiencing higher levels of social isolation. Regarding participants' number of friends, 2.7% reported having none, 2.4% had one friend, 34.0% had 2–3 friends, 50.5% had 4–9 friends, and 10.5% had 10 or more friends. Approximately 5.0% of participants reported not going out. The prevalence of depression was 10.8%, and 24.4% of participants reported poor subjective well-being.

Table 1
Descriptive statistics (N = 2,041)

	n (%)	mean (SD)
Gender		
Men	1074(52.6)	
Women	967(47.4)	
Age		26.2(4.7)
Education		
High school graduate or less	997(48.8)	
Community college graduate	343(16.8)	
Four-year college graduate	701(34.4)	
Income		152.91(139.13)
Marital status		
No spouse	1696(83.1)	
Being married	345(16.9)	
Living arrangements		
Living alone	514(25.2)	
Living with parents or	1527(74.8)	
Living with a partner/spouse		
Economic activity		
Working	1186(58.1)	
Not working	855(41.9)	
Self-rated health		
Very good	1975(1,770)	
Very poor	271(17.28)	
Perceived isolation		3.4(0.84)
Going out		
Going out	1940(95.0)	
Not going out	101(5.0)	

	n (%)	mean (SD)
Number of friends		
0	54(2.65)	
1 friend	49(2.40)	
2–3 friends	694(34.0)	
4–9 friends	1030(50.47)	
10 or more friends	214(10.49)	
Depression		
No	1820(89.2)	
Yes	221(10.8)	
Subjective well-being		
Excellent	1543(75.6)	
Poor	498(24.4)	

Table 2 presents the results of the regression analysis examining the association between perceived isolation, social disconnectedness variables (frequency of going out, number of friends), and depression. The analysis revealed that perceived isolation was significantly associated with depression (OR 1.86, 95% CI = 1.61–2.15), indicating that as perceived isolation increases, young adults are 1.86 times more likely to experience depression. Furthermore, not going out during social isolation was significantly associated with depression (OR 2.59, 95% CI = 1.47–4.58), suggesting that individuals who do not engage in social activities outside their homes are 2.59 times more likely to experience depression. However, the number of friends did not show a significant association with depression, indicating that merely having a higher number of friends does not necessarily protect against depression in young adults. The Hosmer–Lemeshow test indicated a good fit for the model ($p = 0.445$).

Table 2
Relationship between depression and perceived isolation, social disconnectedness (N = 2,041)

Variables	Depression		
	OR	95% CI	p-value
Perceived Isolation	1.86 ***	1.61–2.15	.000
Number of Friends	0.84	0.71–1.01	.060
Frequency of Going Out	2.59 **	1.47–4.58	.001
Age	0.99	0.94–1.04	0.505
Gender	0.84	0.61–1.17	0.370
Marital Status	0.91	0.54–1.54	0.730
Employment	1.36	0.67–2.76	0.396
Income	0.93	0.81–1.07	0.320
Education			
Community College Graduate	1.13	0.71–1.82	.609
Four-Year College Graduate	1.13	0.75–1.69	.565
Family Group Member	0.75	0.52–1.10	.140
Self-Rated Health	0.24***	0.17–0.34	.000
Hosmer–Lemeshow Test: $X^2=1682.43(P = .445)$			
Note: ***p < 0.001, **p < 0.01, *p < 0.05			

Table 3 displays the results of the regression analysis examining the relationship between perceived isolation, social disconnectedness variables (frequency of going out, number of friends), and subjective well-being. The analysis revealed that perceived isolation was significantly associated with subjective well-being (OR 1.17, 95% CI = 1.03–1.33), indicating that as perceived isolation increases, young adults are 1.17 times more likely to have poor subjective well-being. Additionally, not going out during social isolation was significantly associated with subjective well-being (OR 1.96, 95% CI = 1.21–3.16), suggesting that individuals who do not engage in social activities outside their homes are 1.96 times more likely to experience poor subjective well-being. Moreover, the number of friends was significantly associated with subjective well-being (OR 0.79, 95% CI = 0.69–0.90), indicating that a higher number of friends is associated with a 0.21 times lower probability of poor well-being. The Hosmer–Lemeshow test indicated a good fit for the model (p = 0.413).

Table 3
Relationship between subjective well-being and perceived isolation,
social disconnectedness (N = 2,041)

Variables	Subjective Well-Being		
	OR	95% CI	p-value
Perceived Isolation	1.17 *	1.03–1.33	.013
Number of Friends	0.79 **	0.69–0.90	.001
Frequency of Going Out	1.96	1.21–3.16	.006
Age	1.03	1.00–1.07	.056
Gender	1.10	0.88–1.37	.406
Marital Status	0.68	0.48–0.98	.036
Employment	1.16	0.74–1.82	.522
Income	0.94	0.86–1.04	.223
Education			
Community College Graduate	0.90	0.64–1.26	.546
Four-Year College Graduate	1.12	0.85–1.48	.403
Family Group Member	0.96	0.73–1.25	.740
Self-Rated Health	0.23 ***	0.17–0.30	.000
Hosmer–Lemeshow Test: $X^2=1687.08(P = .413)$			
Note: ***p < 0.001, **p < 0.01, *p < 0.05			

Discussion

This study represents a novel examination of the association among perceived isolation, social disconnectedness, and the mental health of 2,041 young adults in South Korea, utilizing data from the 2021 Young Adult Socio-economic Survey. This study offers an analysis of the social connections and the mental health of young adults, a population overlooked in previous research which has primarily focused on older adults or adolescents [48–51]. Notably, this study recognizes that although related, perceived isolation and social disconnectedness are distinct concepts.

The findings indicate that higher levels of perceived isolation were associated with a 1.86 times increased risk of depression and a 1.17 times increased risk of poor subjective well-being, which aligns with the existing research that highlights the substantial impact of perceived isolation on mental health [14, 20].. Regarding social disconnectedness, not going out was associated with a 2.59 times increased risk of

depression and a 1.96 times increased risk of poor subjective well-being. Although the number of friends did not show a significant association with depression, it was significantly associated with an individual's subjective well-being. A higher number of friends was related with a lower probability of poor well-being. These findings align with prior research, indicating that objective social isolation from family and friends was not related with depression and psychological distress. However, it was linked to perceived isolation and an increased presence of depressive symptoms [48]. Also, these results suggest that depression is influenced by a complex interplay of personal, psychological, and social factors [52], and that merely increasing the number of friends may not enable one to overcome depression. Notably, the results from this study exhibit striking parallels with research conducted in the United States, where an investigation focused on the connection between perceived isolation and social disconnection among a cohort of 1,081 young individuals aged 18–35 years. Employing the sophisticated technique of structural equation modeling, the U.S. study unveiled a significant and direct between perceived isolation and adverse mental health outcomes, encompassing depression, anxiety, and the severity of alcohol use [53]. This consistent convergence of findings across geographical contexts underscores the global relevance and urgency of addressing perceived isolation, social connection, as a critical factor influencing mental well-being among young adults. It reinforces the imperative for international awareness and intervention strategies aimed at fostering social connections and promoting mental health within this demographic..

Overall, this study contributes to the existing literature and finds that both perceived isolation and social disconnectedness significantly contribute to poor mental health outcomes, emphasizing their importance as determinants of mental health among Korean young adults and supporting the notion that social connection is an essential factor in mental health [2, 19, 26]. The association observed among perceived isolation, social disconnectedness, and mental health in this study, along with previous research, suggests that interventions aimed at reducing social isolation and promoting social participation among young adults may play a vital role in preventing and addressing mental health issues [49–51]. It is crucial to focus on young adults experiencing high levels of social isolation and explore intervention strategies that emphasize and promote the importance of social connection. The implementation of programs and support systems that encourage the development of close relationships and active social participation among young adults may have significant implications for their mental health.

While this study enhances our understanding of the association among perceived isolation, social disconnectedness, and mental health in the context of young adults in Korea, it has certain limitations. First, the study design relied on cross-sectional survey data, which makes it challenging to establish clear causal relationships. Therefore, caution is warranted when interpreting the findings. Second, the study only assessed a limited aspect of social disconnectedness, as the available secondary data did not allow for a comprehensive examination of various dimensions of social disconnectedness, such as the number of exchanges with family or friends, social support, and social resources. Lastly, this study only focused on depression and subjective well-being as indicators of mental health. However, mental health encompasses a broader range of areas, including stress, anxiety, life satisfaction, and suicidal ideation. Hence, this study is limited by its narrow focus on depression and subjective well-being.

Despite these limitations, as previously emphasized, this study deals with a timely topic considering the high perception of social isolation and poor mental health of young adults in South Korea. The results of this study suggest that the level of social isolation can be considered along with depressive symptoms in young adults, and rapid screening tests can identify groups at high-risk for depression, contributing to efforts to prevent and alleviate depression in young adults. The follow-up study should examine mental health in various areas, such as suicidal thoughts, suicide attempts, and drinking problems.

Conclusions

In conclusion, the findings of this study underscore the critical importance of addressing perceived isolation and social disconnection as key determinants of young adults' mental health. To build upon these insights and contribute to the well-being of the younger population, policymakers and intervention developers should prioritize strategies that not only increase the quantity but also enhance the quality of social relationships [54].

Additionally, it is imperative that interventions focus on reducing loneliness, fostering higher levels of intimacy in interpersonal connections, promoting greater satisfaction with social interactions, and encouraging increased social engagement among young adults. These multifaceted approaches should be integrated into public policies and mental health initiatives aimed at supporting the mental and emotional health of young individuals.

Furthermore, future research should continue to explore innovative interventions and assess their effectiveness in mitigating perceived isolation and social disconnection among young adults. By addressing these factors comprehensively and proactively, we can pave the way for a healthier and more resilient generation, ultimately leading to improved overall mental health outcomes in the young population.

Declarations

Ethics approval and consent to participate

This study did not require institutional review board (IRB) approval or an application for exemption because the data from the Young Adult Socio-economic Survey are secondary data, and anyone can access them. Therefore, there was no need to request IRB approval or an exemption

Consent for publication

Not applicable

Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Author Contributions

All authors made substantial contributions to the conception and drafted this work. Especially, SS made more significant contributions to the analysis and methods as a major contributor this work. All authors read and approved the final manuscript.

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